

Pratyusha Adibhatla

✉ pratyushaadibhatla3@gmail.com ☎ +1 (682) 374-3740 in LinkedIn GitHub Portfolio
H4 EAD(NO SPONSORSHIP REQUIRED) — Current Location: Santa Clara, CA — Relocation: Yes

Summary

Backend Engineer with experience across production APIs, data pipelines, and durable long-running AI workflows. Builds multi-tenant services on Postgres and TypeScript/Python, orchestrates batch ETL/ELT into cloud warehouses, and ships LLM systems with explicit failure handling, retries, and tenant-scoped concurrency control. Comfortable owning a system from idea to deployed and stable in an early-stage, ambiguous environment.

Experience

AI Software Engineer, Finta

May 2026 – Present

- Built durable long-running workflows on Inngest for an LLM document pipeline step level retries with idempotent re-execution, per-org concurrency limits for tenant isolation, and a two-stage generation path (LLM payload to deterministic file spec across PDF/DOCX/CSV/Markdown) running under dual 4.5K/14K character budgets and an 8-minute execution ceiling.
- Shipped student verification end to end in Next.js/Postgres, taking .edu approval from manual SQL to self-serve in under 5 seconds: 6-digit .edu verify with document upload fallback, admin review queue, and a gated Stripe student tier that refuses unverified checkout behind Supabase RLS.
- Made an LLM export pipeline observable and safe to retry. Replaced a lossy JSON fallback with 7 explicit failure paths and upstream tool-error propagation so a failed stage refuses to emit a partial artifact, and split tool errors into model-facing diagnostics vs. sanitized user copy across 8 HTTP status classes, turning silent bad output into structured, traceable failures.
- Shipped 0-to-1 full-stack features across a Python LangGraph agent and a TypeScript/Next.js product: a 7-step activation checklist with agent-assisted CTAs and a 10 touch, 28-day onboarding email engine while scoping an intern's Ollama model-routing work and reviewing teammates' PRs and candidate take-homes.

Data Engineer, UCSD Wireless Communication, Sensing, and Networking Lab

Jan 2025 – Jul 2025

- Orchestrated ETL pipelines using Azure Data Factory (ADF) to trigger Azure Databricks compute, slashing LiDAR storage requirements by 1.2TB (85%) by automating 3D point cloud downsampling and saving optimized Parquet files to Azure Data Lake Storage (ADLS).
- Accelerated query times 75% by architecting a Medallion (Bronze/Silver/Gold) warehouse layer on ADLS with columnar Parquet and lineage from gold tables back to raw sensor source, replacing multi-step manual extraction with analysis-ready DataFrames.
- Eliminated 15+ hours of manual data wrangling per week across the research team by deploying a dynamic categorization engine that parses schema metadata to achieve 100% automated segregation of datasets by sensor type (LiDAR, Radar, WiFi).

Research Assistant - Data Engineer – Retrieval Systems, University at Buffalo

Sep 2024 – Dec 2024

- Established a multimodal vector search pipeline to ingest and structure unstructured data (video, image, text) applying computer vision (OCR) and transcription to convert images and video into retrievable text for low-latency RAG.
- Optimized data indexing workflows in ChromaDB by designing metadata schema, regenerating embeddings, and refreshing vector indexes when new data was added to ensure consistency across modalities.

Associate Data Engineer, Cognizant (Client: Liberty Mutual Insurance)

Dec 2020 - Jun 2022

- Optimized batch ETL on AWS EC2 processing 1M+ monthly transactions; carried on-call for pipeline health and SLA adherence, triaging 20+ weekly failures and re-running failed stages without duplicating downstream records. Automated build, test, and deploy for ETL workflows with Jenkins and Git, cutting manual release effort 25%.
- Orchestrated warehouse SQL transformations across Amazon RDS and Redshift, resolving cross-system eventual consistency gaps and stranded payment records using DynamoDB metadata as the reconciliation ledger.
- Implemented automated data quality validation for schema drift, null anomalies, and reconciliation inconsistencies, reducing data errors 26% across audit-compliant financial transactions.

Technologies

Languages: Python, TypeScript, SQL, Java · React, Next.js

Backend & Infra: REST APIs (FastAPI/Flask, Next.js), SSE streaming, OAuth2, RBAC, multi-tenant isolation (Postgres RLS), durable workflows (Inngest), retries/idempotency/concurrency control, AWS (EC2/S3/RDS/Redshift/DynamoDB), Azure (ADF/Databricks/ADLS), Docker, CI/CD (Jenkins, GitHub Actions), Vercel, structured logging & incident triage

AI & Data: LangGraph agentic workflows, RAG, structured extraction, hybrid retrieval/RRF, batch & streaming ETL/ELT, data modeling, schema-drift validation, lineage, Parquet, warehouses (Redshift, Snowflake, BigQuery), PySpark, Pandas, Postgres/Supabase/pgvector, ChromaDB

Familiar: Snowflake, BigQuery

Education

Masters in Data Science | University of Texas at Arlington Aug 2022 – May 2024

- Relevant Coursework: Machine Learning, Data Science, Data Analytics, Database Systems, Data Mining, Neural Networks, Probability and Statistics.

BTech in ECE | Andhra University Visakhapatnam, India Jun 2016 – Sep 2020